

Mathematical Model of Ranking Accuracy and Popularity Promotion

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Abstract—Elo rating system has been widely used to predict the outcomes in various domains such as board games and sports games. However, Fédération Internationale de Football Association (FIFA) like other associations use their own ranking system with certain purposes other than ranking accuracy only. This paper discuss studies of four different ranking methods: Elo rating system, Betting Odds, FIFA rating based on Elo style and FIFA rating based on Davidson Wang's method, with which 36 round-robin matches of UEFA Euro 2016 are analyzed. We then explore entertainment aspects based on uncertainty of outcome prediction that may have occurred by ranking methods. Moreover, this paper considers the trade-off between the ranking accuracy and popularity promotion. Further investigation is needed to assess the well balance between ranking accuracy and popularity promotion.

Keywords: *FIFA ranking system, Betting Odds, Elo rating system, UEFA Euro 2016*

I. INTRODUCTION

Elo rating system has been widely used to predict the outcome in various domains such as board games and sports games [5]. This is because Elo rating system is statistically accurate and stable. Note that the phrase “Elo rating” is often used to mean a player's chess rating as calculated by the World Chess Federation (FIDE). However, this usage is confusing and misleading, because Elo's general ideas have been adopted by many organizations such as Fédération Internationale de Football Association (FIFA). Interestingly each organization has a unique implementation, and none of them follows Elo's original suggestions precisely. This study concerns the difference between Elo rating and other ranking systems from the viewpoint of ranking accuracy and popularity promotion. In fact, little is known about its potential effect to popularity promotion. To the best of our knowledge, this study is the first serious observation in this direction.

Elo rating system was initially developed for assessing the strength of chess players [5], but has been widely adopted in various sports including association football [3], then the Elo style method has been used to predict the outcome of the Premier League [10]. On the other hand, a certain ranking system has been established in a community or association.

For example, FIFA like other associations established their own ranking system.

Moreover, Dan ranking system is used by many Japanese organizations and Korean martial arts to indicate the level of one's ability within a certain subject matter. As a ranking system, it was originally used at a Go school during the Edo period [8]. It is now also used in modern fine arts and martial arts. The system was applied to martial arts in Japan by Kano Jigoro [9], the founder of Judo, in 1883, and later introduced to other East Asian countries. In the modern Japanese martial arts, holder of Dan ranks often wear a black belt. Dan ranks are also given for board games such as Go, Japanese chess (Shogi), and Renju, as well as for cultural arts such as flower arrangement (Ikebana), Japanese calligraphy (Shodo) and tea ceremony (Sado).

These examples may suggest the importance of a good balance between ranking accuracy and popularity promotion in the domains considered.

In this study, soccer or association football was chosen as a benchmark. The UEFA Euro 2016 tournament is hosted by France, where 24 teams play in this tournament. For these 24 teams, we analyze the winning percentage for each match in round-robin using different ranking methods: Elo rating system, Betting Odds, FIFA rating based on Elo style and FIFA rating based on Davidson Wang's method. We then discuss the ranking accuracy and its potential influence to popularity promotion.

The structure of this paper is as follows. Section II presents four different outcome prediction methods. Section III explains the detail of each method and compares. Section IV discusses the results of our analysis. Section V gives concluding remarks.

II. OUTCOME PREDICTION MODELS

In this section, four types of mathematical models for the outcome prediction are presented, and the winning probability outcomes are compared between them.

A. Official FIFA Ranking System

FIFA World Ranking is a ranking system for men's national teams in association football. The teams of the member nations

of FIFA are ranked based on their game results with the most successful teams being ranked highest [2]. This ranking was introduced in December 1992 and most recently revamped after 2006 World Cup, with the first edition of the new series of rankings issued on 12 July 2006. In FIFA ranking system, each team has its own rating point, which is used to determine the ranking position in the full team list.

The rating point refers to the last 48 months scores with weightage of 0.2, 0.3, 0.5 and 1 for every 12 months. For a chosen team the formula for the calculation of points P awarded after a single game is given by Equation (1),

$$P = M * I * T * C * 100 \quad (1)$$

where each parameter has the following meaning.

- M : the outcome (win=3, draw=1 and lose=0)
- I : the match importance (see Table I)
- T : the strength of opposition (see Equation (2))
- C : the confederation strengths (see Table II)

Table I
MEANING OF I : MATCH IMPORTANCE MULTIPLIERS

Competition	I
World Cup final competition	4.0
Continental final competition and FIFA Confederations Cup	3.0
World Cup qualifier and continental-level qualifier	2.5
Friendly match (including small tournaments)	1.0

Strength of opposition T is calculated according to the following formula:

$$T = \frac{200 - r}{100} \quad (2)$$

where r stands for the ranking position of the opposing team. Only the team at the top of the ranking is assigned the value near 2.00 (when $r = 1$), teams ranked 150th and below are assigned the minimum weightage of $T = 0.5$.

The confederation strength C is a dynamic parameter calculated on the basis of the number of victories by confederation at the last FIFA World Cup competitions. The new values are shown in Table II. Once we calculate the points that a team earned

Table II
MEANING OF C : CONFEDERATION VALUES

Confederation	C
Confederación Sudamericana de Fútbol	1.00
Union of European Football Associations	0.99
Asian Football Confederation	0.85
Confederation of African Football	0.85
Oceania Football Confederation	0.85
CONCACAF	0.85

in one match, we can obtain the whole points over a period of 4 years (48 months). In the consecutive years the mean of the accumulated points is computed, then the 4 yearly averages are summed up with weights 0.2, 0.3, 0.5 and 1. Finally, FIFA can compute each teams' scores and publish the new FIFA rating. In fact, until now there is no good mathematical model to predict the game outcome by FIFA ranking system. Therefore, in Section III, we introduce two measurements to calculate the winning probability based on FIFA rating [17].

B. Elo Rating System in Soccer

Elo rating system is a method for calculating the relative skill levels of players. It was created by Arpad E. Elo in 1960 originally for chess rating system and was approved at USCF (United States Chess Federation) in that year. Later, In 1970 it was agreed by FIDE [5]. The World Football Elo Ratings [5] is based on the Elo rating system but modified to take various football-specific variables into account. The rating points are calculated by the following formula:

$$R_n = R_o + KG(W - W_e) \quad (3)$$

where R_n and R_o is the new and old team rating point, respectively. K is a dynamic weightage index regarding the tournament of the match. Similar to the FIFA rating system, the different tournament level has different K value, as shown in Table III.

Table III
MEANING OF K : TOURNAMENT OR MATCH TYPE

Tournament or Match type	K
World Cup	60
Continental Championship and Intercontinental Tournaments	50
World Cup and Continental qualifiers and major tournaments	40
All other tournaments	30
Friendly Matches	20

G denotes the number of goals. If the game is a draw or is won by one goal then $G = 1$. If the game is won by two goals then $G = 1.5$. If the game is won by three or more goals the formula $G = \frac{11+N}{8}$ is applied.

W and W_e represents the match outcome ($W = 1$ for win, $W = 0.5$ for draw, and $W = 0$ for lose) and the winning expectation, respectively. While two teams have a competition, the winning expectation of team a (say W_{ea}) is given by Equation (4),

$$W_{ea} = \frac{1}{1 + 10^{\frac{-dr}{400}}} \quad (4)$$

where dr stands for the difference in ratings (add 80 points for the home team).

We show, in Figure 1, the distribution of expected results. The upper (lower) line shows the expected result when the opponent is in lower (higher) ranking by Elo rating measurement. When two teams are in the same ranking, the expect result is 0.5 (draw). When two teams have a little gap in ranking, the expected result will increase or decrease rapidly. But when two teams have a big gap in ranking, the expected result will increase or decrease slowly. This means that the changing values of expected result get smaller per gap which is opposite to realities.

The original Elo rating measurement was used to analyze the winning percentage of board game that have rare draw games, however for soccer there are a lot of draw outcomes. Therefore, we need to evaluate and infer the function in [10], the draw probability $P(\text{draw elo})$ which fits the normal distribution is shown in Equation (5),

$$P(\text{draw elo}) = f(x) = \frac{1}{\sqrt{2\pi e}} \exp\left(-\frac{\left(\frac{dr}{200}\right)^2}{2e^2}\right) \quad (5)$$

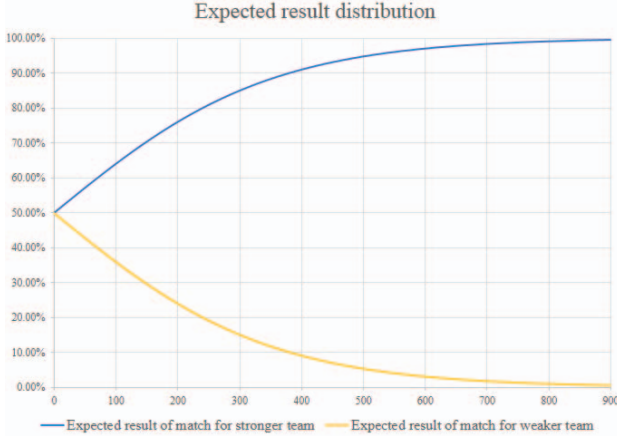


Figure 1. The distribution of expected results W_e by Elo rating measurement

where e is Euler's number. Then, the probability of win and lose is given by Equations (6) and (7), respectively.

$$P(\text{win elo}) = \frac{1}{1 + 10^{\frac{-dr}{400}}} - \frac{1}{2}P(\text{draw elo}) \quad (6)$$

$$P(\text{lose elo}) = \frac{1}{1 + 10^{\frac{dr}{400}}} - \frac{1}{2}P(\text{draw elo}) \quad (7)$$

Then, the game outcomes are depicted as Figure 2.

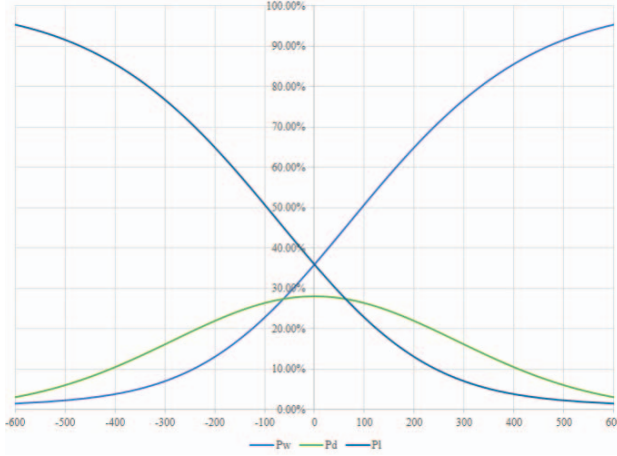


Figure 2. Probability distributions of win, draw and lose by Elo rating measurement

C. Betting Odds System

Another useful prediction method is Betting Odds [16]. Odds are a numerical expression, usually expressed as a pair of numbers, used in both gambling and statistics [15]. In statistics, the odds for or odds of some event reflect the likelihood that the event will take place. Odds against reflect the likelihood that a particular event will not take place. In gambling, the odds are the ratio of payoff to stake, and do not necessarily reflect the exact probabilities. Odds are expressed in several ways, and sometimes the term is used incorrectly to mean simply the probability of an event. Conventionally, gambling

odds are expressed in the form "X to Y", where X and Y are numbers, and it is implied that the odds are odds against the event on which the gambler is considering wagering.

In both gambling and statistics, the 'odds' are a numerical expression of the likelihood of some possible event. In gambling, odds represent the ratio between the amounts staked by parties to a wager or bet. Thus, odds of 6 to 1 mean that the first party stakes six times the amount staked by the second party. We can get an easy example to explain how betting odds comes: while *Team A* vs. *Team B*, 1,000 audience predict the result and 900 audience choose *Team A* win; 90 audience choose draw game and 10 audience choose *Team B* win. Then we have odds for *Team A* as $O(w) = \frac{1000}{900} = 1.11$, $O(d) = \frac{1000}{90} = 11.1$ and $O(l) = \frac{1000}{10} = 100$. In order to earn money, betting company will decrease the odds by revising as (1.06, 10.3, 90).

In this study, we focus on the individual match of UEFA Euro 2016 round-robin tournament and employ betting odds data to generate a measure of uncertainty of outcome by Ladbrokes company [12]. Based on the odds ($O(w), O(d), O(l)$) we can obtain the probability [15] by Equations (8) - (10).

$$P(\text{win bet}) = \frac{1}{\frac{1}{O(w)} + \frac{1}{O(d)} + \frac{1}{O(l)}} * \frac{1}{O(w)} \quad (8)$$

$$P(\text{draw bet}) = \frac{1}{\frac{1}{O(w)} + \frac{1}{O(d)} + \frac{1}{O(l)}} * \frac{1}{O(d)} \quad (9)$$

$$P(\text{lose bet}) = \frac{1}{\frac{1}{O(w)} + \frac{1}{O(d)} + \frac{1}{O(l)}} * \frac{1}{O(l)} \quad (10)$$

III. EXPLANATION AND COMPARISON

In this section, Wang's [19] and Davidson's [4] model-based ideas are introduced. As mentioned previously, we need to model the probability of ties in a soccer match which is not available in the standard Bradley-Terry model. There are various ways for constructing an extended Bradley-Terry model in the literature. Then in Wang's work [19] models the tie probability similarly to Davidson [4]. The basic idea is derived from regularly occurring patterns in the football world. The closer in strength the two playing teams are, the more likely is a draw [18].

In contrast, if there is a relatively huge strength difference between the two playing teams, it is very likely that the stronger team will win the game. With this in mind, $P(\text{win Davidson})$ which means the probability that team 1 wins; $P(\text{draw Davidson})$, the probability that game ends with a tie and $P(\text{lose Davidson})$, the probability that team 2 wins game can be calculated by the following Equations (11) - (13),

$$P(\text{win Davidson}) = \frac{\beta_1}{\beta_1 + \beta_2 + K\sqrt{\beta_1\beta_2}} \quad (11)$$

$$P(\text{draw Davidson}) = \frac{K\sqrt{\beta_1\beta_2}}{\beta_1 + \beta_2 + K\sqrt{\beta_1\beta_2}} \quad (12)$$

$$P(\text{lose Davidson}) = \frac{\beta_2}{\beta_1 + \beta_2 + K\sqrt{\beta_1\beta_2}} \quad (13)$$

where β_1 and β_2 denote the strength parameters of the first and second team in game respectively. K is the tie effect to be estimated as 0.905 [19].

We use the FIFA rating to present the strength for each team. Another measurement is “Elo Style”; use the same functions as Equations (5) - (7), just replace the Elo rating to FIFA rating. The FIFA rating data of UEFA Euro 2016 is shown in Table IV (*Data from date: 2nd May, 2016*) [7]. Similarly, we have the Elo rating data of UEFA Euro 2016 as shown in Table V, since France is the host, we add 80 points to France (*Data from date: 2nd, May, 2016*) [6].

Then we calculate the round-robin match of UEFA Euro 2016, for each match a number was given. We show, in Table VI, the 36 rounds of game prediction for win, draw and lose. Using Equations (8) - (10), we obtain Table VI which shows the 36 rounds of game prediction for win, draw and lose. The data comes from company Ladbrokes on 2nd-May-2016 [12]. Finally, based on FIFA rating, we use David-Wang and “Elo Style” methods according to Equations (5) - (7) and Equations (11) - (13) to obtain Table VIII.

In order to analyze the data, we put the outcomes of winning probability by four prediction methods together to draw the poly-line as shown in Figure 3. Similarly, we show, in Figure 4

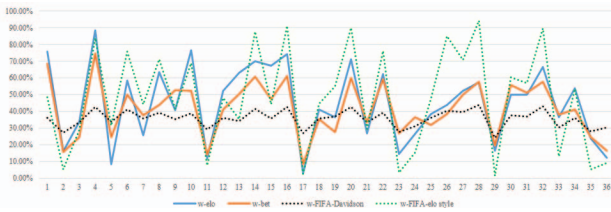


Figure 3. Winning probability by four prediction methods

and Figure 5, draw and losing probability by using the four prediction methods, respectively.

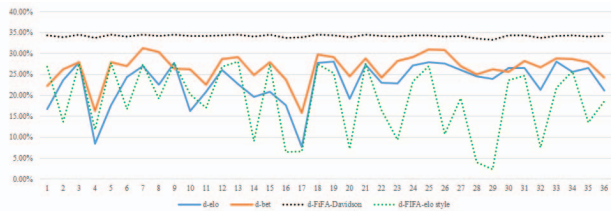


Figure 4. Draw probability by four prediction methods

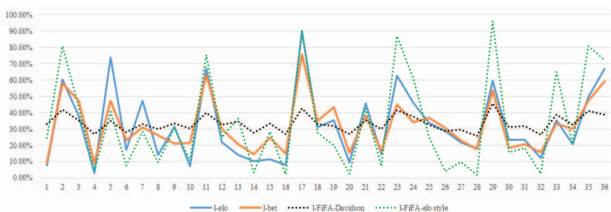


Figure 5. Losing probability by four prediction methods

IV. DISCUSSION

In this section we focus on ranking accuracy and its potential influence to popularity promotion to be discussed based on the data analysis described previously. Figures 3 - 5 clearly show that Elo rating system is better than FIFA ranking system from the viewpoint of ranking accuracy. Hence we focus on the other aspects to be considered.

A. Benefits and drawbacks of FIFA ranking system

FIFA ranking does not only evaluate the relative strength of all 209 international teams, but is also useful when categorizing the teams within a continent into different points. This is especially helpful for the draws in the FIFA World Cup qualifiers or continental championships [19].

FIFA ranking system is a points-earning system. Theoretically, two teams both ranked 150th or below can always benefit from playing a large number of friendly games against each other and negotiate to win half of these games each [13]. They can apply this strategy without any cost.

Although the points-based FIFA ranking is easy to operate and modify, it is not obvious how the points should be used to predict the 3 possible outcomes in a game. As the average performance difference between home and away games for almost all football teams has been revealed by various researchers, a team can benefit more from playing a higher percentage of its games at home. A more realistic method should take the home effect into account such that the ranking can be determined by the pure strength parameters of the Elo rating system [19].

FIFA ranking currently assigns a fixed depreciation weight for all games played within a 12-month period. Although the basic idea is that the older the game is, the lower the weight should be assigned to it, its depreciation function is piecewise with 3 discontinuous points, which seems not very reasonable [13].

B. Potential Influence to Popularity Promotion

When introducing a ranking system in a target domain, what is the most important aspect to be considered? Ranking accuracy should be important to maintain fairness, but may be not the most, or it depends on the domain or the current situation. In a sense, having an accurate ranking system may imply that only few people (winners) would be happy while many others are not so. It would affect the popularity promotion in the domain considered. FIFA ranking system also follows this way, increasing uncertainty of outcome prediction and entertaining more people are two significant ways to promote the popularity of football.

FIFA ranking is used to choose the seeds teams in World Cup qualification in each confederation according to their scores. So it is very important to increase the uncertainty in ranking to attract more people. A significant approach is to use the strength of opposite “ T ” instead of expected results “ W ” in Equation (1). For example, the minimum score condition when one team wins one match is: Friendly Match; The strength of opposite T is 0.5; Confederation score is 0.85. So the minimum score one

Table IV
FIFA RATING OF UEFA EURO 2016

Group A & B	Rating	Group C & D	Rating	Group E & F	Rating
France	1007	Poland	821	Ireland	792
Albania	632	Germany	1309	Belgium	1352
Romania	922	Ukraine	880	Italy	959
Switzerland	974	Northern Ireland	825	Sweden	713
Wales	839	Turkey	983	Austria	1067
England	1069	Spain	1277	Portugal	1184
Russia	821	Czech	810	Iceland	724
Slovakia	784	Croatia	856	Hungary	925

Table V
ELO RATING OF UEFA EURO 2016

Group A & B	Rating	Group C & D	Rating	Group E & F	Rating
France	1947+80	Poland	1762	Ireland	1751
Albania	1578	Germany	2037	Belgium	1896
Romania	1738	Ukraine	1797	Italy	1844
Switzerland	1753	Northern Ireland	1578	Sweden	1735
Wales	1638	Turkey	1797	Austria	1766
England	1934	Spain	1977	Portugal	1885
Russia	1747	Czech	1728	Iceland	1647
Slovakia	1715	Croatia	1794	Hungary	1671

Table VI
ELO RATING AND WINING PROBABILITY OF UEFA EURO 2016

No.	Home Team	Away Team	dr	P(w-elo)	P(d-elo)	P(l-elo)
1	France	Romania	289	75.70%	16.74%	7.56%
2	Albania	Switzerland	-166	15.96%	23.63%	60.41%
3	Romania	Switzerland	-15	33.86%	27.96%	38.18%
4	France	Albania	440	88.39%	8.50%	3.11%
5	Switzerland	France	-274	8.30%	17.64%	74.06%
6	Romania	Albania	151	58.29%	24.33%	17.38%
7	Wales	Slovakia	-77	25.60%	27.00%	47.41%
8	England	Russia	187	63.29%	22.58%	14.13%
9	Russia	Slovakia	32	40.68%	27.82%	31.50%
10	England	Wales	296	76.44%	16.33%	7.23%
11	Slovakia	England	-219	11.67%	20.84%	67.49%
12	Russia	Wales	109	52.18%	26.02%	21.80%
13	Poland	Northern Ireland	184	62.89%	22.73%	14.38%
14	Germany	Ukraine	240	70.10%	19.64%	10.26%
15	Ukraine	Northern Ireland	219	67.49%	20.84%	11.67%
16	Germany	Poland	275	74.18%	17.58%	8.25%
17	Northern Ireland	Germany	-459	2.82%	7.65%	89.53%
18	Ukraine	Poland	35	41.13%	27.79%	31.09%
19	Turkey	Croatia	3	36.43%	28.00%	35.57%
20	Spain	Czech	249	71.19%	19.11%	9.70%
21	Czech	Croatia	-66	26.99%	27.26%	45.76%
22	Spain	Turkey	180	62.34%	22.94%	14.72%
23	Croatia	Spain	-183	14.47%	22.78%	62.75%
24	Czech	Turkey	-69	26.60%	27.19%	46.21%
25	Ireland	Sweden	16	38.32%	27.96%	33.72%
26	Belgium	Italy	52	43.66%	27.54%	28.80%
27	Italy	Sweden	109	52.18%	26.02%	21.80%
28	Belgium	Ireland	145	57.43%	24.60%	17.97%
29	Sweden	Belgium	-161	16.42%	23.87%	59.71%
30	Italy	Ireland	93	49.80%	26.55%	23.65%
31	Austria	Hungary	95	50.10%	26.49%	23.42%
32	Portugal	Iceland	211	66.47%	21.29%	12.25%
33	Iceland	Hungary	3	36.43%	28.00%	35.57%
34	Portugal	Austria	119	53.65%	25.66%	20.68%
35	Iceland	Austria	-92	23.77%	26.58%	49.65%
36	Hungary	Portugal	-214	12.03%	21.12%	66.85%

team wins the match can obtain $3 * 1 * 0.5 * 0.85 * 100 = 127.5$ points. But when using the Elo rating system, when two teams have big strength gap the minimum score of winning is close to

0. Therefore, the official FIFA rating can speed up the changing of rating and create more uncertainty to attract fans to pay more attention to football game.

Table VII
BETTING ODDS AND WINING PROBABILITY OF UEFA EURO 2016

No.	Home Team	Away Team	Odd(w)	Odd(d)	Odd(l)	P(w-bet)	P(d-bet)	P(l-bet)
1	France	Romania	1.36	4.2	10	68.50%	22.18%	9.32%
2	Albania	Switzerland	6	3.6	1.62	15.70%	26.16%	58.14%
3	Romania	Switzerland	3.85	3.4	2	24.65%	27.91%	47.44%
4	France	Alabania	1.27	5.8	10.5	74.63%	16.34%	9.03%
5	Switzerland	France	3.85	3.4	2	24.65%	27.91%	47.44%
6	Romania	Albania	1.9	3.5	4.1	49.84%	27.06%	23.10%
7	Wales	Slovakia	2.5	3	3	37.50%	31.25%	31.25%
8	England	Russia	2.15	3.1	3.6	43.65%	30.28%	26.07%
9	Russia	Slovakia	1.8	3.6	4.5	52.63%	26.32%	21.05%
10	England	Wales	1.75	3.5	4.25	52.31%	26.15%	21.54%
11	Slovakia	England	6.7	4.2	1.5	14.16%	22.59%	63.25%
12	Russia	Wales	2.3	3.3	3.15	41.20%	28.72%	30.08%
13	Poland	Northern Ireland	1.85	3.2	4.5	50.27%	29.06%	20.67%
14	Germany	Ukraine	1.55	3.8	6.5	60.74%	24.78%	14.48%
15	Ukraine	Northern Ireland	2	3.4	3.85	47.44%	27.91%	24.65%
16	Germany	Poland	1.55	4	6.3	61.22%	23.72%	15.06%
17	Northern Ireland	Germany	11.5	6	1.25	8.25%	15.82%	75.93%
18	Ukraine	Poland	2.7	3.2	2.7	35.16%	29.67%	35.16%
19	Turkey	Croatia	3.4	3.2	2.15	27.44%	29.16%	43.40%
20	Spain	Czech	1.55	3.8	6	60.02%	24.48%	15.50%
21	Czech	Croatia	2.85	3.3	2.5	33.29%	28.75%	37.95%
22	Spain	Turkey	1.6	3.9	5.7	59.14%	24.26%	16.60%
23	Croatia	Spain	3.55	3.35	2.1	26.67%	28.26%	45.08%
24	Czech	Turkey	2.6	3.25	2.75	36.42%	29.14%	34.44%
25	Ireland	Sweden	2.9	3	2.5	31.98%	30.92%	37.10%
26	Belgium	Italy	2.4	3	3	38.46%	30.77%	30.77%
27	Italy	Sweden	1.9	3.5	4.1	49.84%	27.06%	23.10%
28	Belgium	Ireland	1.65	3.8	5.4	57.48%	24.96%	17.56%
29	Sweden	Belgium	4.8	3.6	1.75	19.70%	26.27%	54.03%
30	Italy	Ireland	1.7	3.7	5.1	55.78%	25.63%	18.59%
31	Austria	Hungary	1.83	3.3	4.5	50.99%	28.28%	20.74%
32	Portugal	Iceland	1.62	3.5	6	57.71%	26.71%	15.58%
33	Iceland	Hungary	2.5	3.3	2.85	37.95%	28.75%	33.29%
34	Portugal	Austria	2.3	3.3	3.15	41.20%	28.72%	30.08%
35	Iceland	Austria	3.85	3.4	2	24.65%	27.91%	47.44%
36	Hungary	Portugal	5.7	3.9	1.6	16.60%	24.26%	59.14%

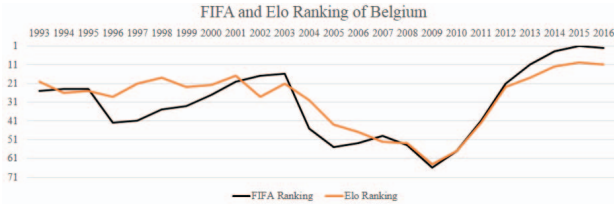


Figure 6. Belgium's FIFA ranking and Elo-rating-based ranking compared

Figure 6 shows that FIFA ranking of Belgium Men's National Football Team is No.71 in 2007, whereas after 7 years Belgium has raised to No.1 in 2015 (October 15th). However, the Elo ranking seems more smooth and provides less excitement for audiences.

On the other hand, the FIFA rating has a higher possibility of draw for two teams with a wider gap. It can attract more people in disadvantaged national team's fans to perceive that their team will not lose the game. Therefore, the population of football fans increase rapidly, especially in the disadvantaged nation area.

C. Balance between Skill and Chance

The balance between skill and chance is an important issue from the entertainment's point of view. An interesting game should not depend too much on skill or chance, but should have a harmonic balance which provides comfortable uncertainty of game outcome prediction (See Figure 7). This idea has been studied in the framework of game refinement theory [11]. A measurement to quantify game sophistication (i.e., balance between skill and chance) has been proposed based on the game outcome uncertainty in the sense that the game outcome should be uncertain until the very end of game.

Assuming that ranking system is a kind of game playing (from the organizer's point of view), a good balance between skill and chance should be examined. In the domain of ranking system, ranking accuracy may correspond to the skill part, whereas uncertainty of ranking may correspond to the chance part. Roughly speaking, Elo rating system is related to player's skills, while FIFA ranking system is located in the level somewhere between skill and chance.

Table IX shows the difference between the two ranking systems. From the table we know FIFA ranking system can provide more unexpected results for audience to attract more population attend the game. On the other hand, Elo rating

Table VIII
FIFA RATING, WINING PROBABILITY BY DAVIDSON METHOD AND ELO RATING STYLE OF UEFA EURO 2016

No.	Home Team	Away Team	P(w-dav)	P(d-dav)	P(l-dav)	P(w-elos)	P(d-elos)	P(l-elos)
1	France	Romania	35.95%	34.40%	32.92%	48.60%	26.78%	24.62%
2	Albania	Switzerland	27.29%	33.88%	42.05%	5.44%	13.63%	80.93%
3	Romania	Switzerland	33.48%	34.41%	35.37%	28.80%	27.54%	43.66%
4	France	Alabania	42.65%	33.79%	26.77%	83.76%	11.78%	4.46%
5	Switzerland	France	33.85%	34.42%	35.00%	31.36%	27.81%	40.83%
6	Romania	Albania	41.07%	34.00%	28.15%	75.81%	16.68%	7.51%
7	Wales	Slovakia	35.60%	34.41%	33.26%	44.11%	27.48%	28.41%
8	England	Russia	39.05%	34.22%	29.99%	71.07%	19.17%	9.76%
9	Russia	Slovakia	35.22%	34.42%	33.63%	41.42%	27.76%	30.81%
10	England	Wales	38.66%	34.25%	30.34%	68.88%	20.22%	10.91%
11	Slovakia	England	29.24%	34.14%	39.87%	7.75%	16.98%	75.27%
12	Russia	Wales	36.04%	34.40%	32.83%	48.15%	26.86%	24.98%
13	Poland	Northern Ireland	34.34%	34.42%	34.51%	35.43%	28.00%	36.58%
14	Germany	Ukraine	41.42%	33.96%	27.85%	87.69%	9.02%	3.29%
15	Ukraine	Northern Ireland	35.54%	34.41%	33.32%	44.11%	27.48%	28.41%
16	Germany	Poland	42.66%	33.79%	26.76%	91.09%	6.46%	2.45%
17	Northern Ireland	Germany	26.83%	33.80%	42.58%	2.50%	6.62%	90.88%
18	Ukraine	Poland	35.62%	34.41%	33.24%	44.71%	27.41%	27.89%
19	Turkey	Croatia	36.83%	34.37%	32.07%	54.83%	25.35%	19.82%
20	Spain	Czech	42.46%	33.82%	26.93%	89.98%	7.31%	2.71%
21	Czech	Croatia	33.48%	34.41%	35.38%	29.60%	27.64%	42.76%
22	Spain	Turkey	39.00%	34.22%	30.02%	76.23%	16.44%	7.32%
23	Croatia	Spain	27.80%	33.95%	41.47%	3.44%	9.40%	87.16%
24	Czech	Turkey	31.15%	34.31%	37.80%	15.33%	23.29%	61.38%
25	Ireland	Sweden	36.25%	34.39%	32.63%	47.70%	26.94%	25.35%
26	Belgium	Italy	40.46%	34.08%	28.70%	85.16%	10.82%	4.02%
27	Italy	Sweden	39.62%	34.16%	29.46%	70.83%	19.29%	9.88%
28	Belgium	Ireland	43.89%	33.59%	25.71%	94.14%	4.06%	1.80%
29	Sweden	Belgium	24.14%	33.24%	45.78%	1.33%	2.27%	96.40%
30	Italy	Ireland	37.76%	34.32%	31.18%	60.55%	23.58%	15.87%
31	Austria	Hungary	36.91%	34.36%	31.99%	57.00%	24.73%	18.27%
32	Portugal	Iceland	43.12%	33.72%	26.37%	89.58%	7.61%	2.81%
33	Iceland	Hungary	30.30%	34.25%	38.71%	13.00%	21.83%	65.16%
34	Portugal	Austria	36.23%	34.39%	32.65%	53.36%	25.74%	20.90%
35	Iceland	Austria	27.99%	33.98%	41.25%	5.41%	13.57%	81.02%
36	Hungary	Portugal	30.27%	34.24%	38.74%	9.12%	18.53%	72.36%

Table IX
MAIN DIFFERENT POINTS OF FIFA AND ELO RANKING

	FIFA ranking	Elo ranking
1	Only used in FIFA soccer	Commonly used in variety of games
2	Have no effective math model to predict result	Predict game outcome accurately
3	Mutability	Stable
4	Big gap rating	Small gap rating

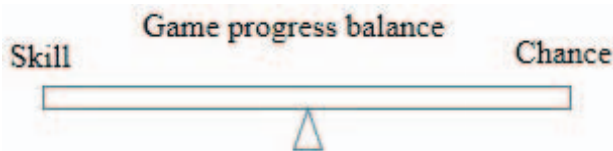


Figure 7. Illustration to show a balance between skill and chance in games

system can always show the actual ranking which means that the system focus more on skill and less on chance, then FIFA refuse to use it.

In fact, not only FIFA, but also Shogi and Go professional communities do not use Elo rating system. Instead they use “Dan System”, in which, while a player gets 9 Dan in 30 years old, he will keep the status forever. However, in Elo rating system, all players must change their rankings by their records

and strength. Therefore, in order to protect player’s self-respect and attract more people who prefer to enjoy the game, Elo rating system was not employed. It means that if we want to make a game become more popular, Elo rating system is not a good choice, at least it is better not to show explicitly. Elo rating system just focus on the real strength and no other things to be covered, therefore if we want to promote the target game, Elo rating system is not a good choice.

Interestingly, FIFA uses Elo rating system for Women’s football. We notice that FIFA chooses the different popularity promotion strategies for Men and Women games. Therefore, FIFA also understands that Elo rating system is quite useful and accurate. For the phenomenon, compared with Men’s football, Women’s football is currently not so popular, FIFA can perform an experiment of Elo rating system.

D. Game Refinement Theory and Ranking System

In this section, we will guide game refinement theory to analyze two ranking system by mathematical way[20]. Game information progress presents how certain is the result of the game in a certain time or steps. Let F and T be the average number of ranking float and the time period, respectively. If one knows the game information progress, for example after the game, the game progress $x(t)$ will be given as a linear function of time t with $0 \leq t \leq T$ and $0 \leq x(t) \leq F$, as shown in Equation (14).

$$x(t) = \frac{F}{T} t \quad (14)$$

However, the game information progress given by Equation (14) is usually unknown during the in-game period. Hence, the game information progress is reasonably assumed to be exponential. This is because the game outcome is uncertain until the very end of game in many games. Hence, a realistic model of game information progress is given by Equation (15).

$$x(t) = F\left(\frac{t}{T}\right)^n \quad (15)$$

Here n stands for a constant parameter which is given based on the perspective of an observer in the game considered. Then acceleration of game information progress is obtained by deriving Equation (15) twice. Solving it at $t = T$, the equation becomes

$$x''(T) = \frac{F n(n-1)}{T^n} t^{n-2} = \frac{F}{T^2} n(n-1)$$

It is assumed in the current model that the game information progress in any type of games is happening in our minds. We do not know yet about the physics in our minds, but it is likely that the acceleration of information progress is related to the force in mind. Hence, it is reasonably expected that the larger the value $\frac{F}{T^2}$ is, the more the game becomes exciting due to the uncertainty of game outcome. Thus, we use its root square, $\frac{\sqrt{F}}{T}$, as a game refinement measure for the game considered.

According to the Figure 6, we can know the ranking position in each year for Belgium. First we calculate the average ranking range for two ranking systems. The data from 1993 year to 2016 year and define the ranking position in 1993 is R_0 and ranking position in 2016 is R_{23} , then the average ranking range should be

$$R_{avg} = \frac{\sum_{i=0}^{22} |R_{i+1} - R_i|}{23}$$

Then we calculate the parameter F , the average ranking float is

$$F = \frac{\sum_{i=0}^{22} ||R_{i+1} - R_i| - R_{avg}|}{23}$$

Final we get $F_{fifa}=5.55$ and $F_{elo}=3.60$ to calculate $R_{FIFA} = \frac{\sqrt{F}}{T} = 0.102$ and $R_{Elo} = \frac{\sqrt{F}}{T} = 0.082$. In game refinement theory, the higher R value means the higher entertainment with low competition. Therefore, FIFA ranking system can attract more people to join game.

V. CONCLUSION

In this paper, comparing between the four mathematical models (Elo rating system, betting odds system, FIFA rating based on Davidson Wang's method, and FIFA rating based on Elo rating style) we have found the advantages of each measurement. FIFA ranking system can provide more chance and interesting points for the supporter or fans. The most accurate measurement is Elo rating system, however many associations and organizations refuse to use it although it is accurate. Thus FIFA ranking is very useful to attract more people to join the game and keep the enthusiasm of fans. In addition, we now know how to attract more people who prefer to play the target game. Also in this paper, we have inferred and cracked the draw probability function in Elo rating system.

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